

Automated Long-Answer Grading using QLoRA Fine-Tuned Qwen2.5-3B-Instruct

Summer Internship Project Report

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Abstract

Evaluating descriptive or long-form student answers is a time-consuming and subjective task for educators. This project develops an automated grading system for long handwritten-style answers using a large language model (LLM). The base model is Qwen2.5-3B-Instruct, an instruction-following language model with approximately three billion parameters. To adapt it to the grading task without requiring expensive full-parameter fine-tuning, Quantized Low-Rank Adaptation (QLoRA) is used: the model is loaded in 4-bit NF4 precision and trained with lightweight LoRA adapters attached to the attention and MLP projection layers.

The dataset consists of 1,000 student answer records covering 10 questions from computer science and data engineering topics. Each record includes the question, student answer, a rubric with up to 10 scored criteria, and a final human-assigned score. The dataset was split into 700 training, 150 validation, and 150 held-out test records. A separate new external test set of 1,000 records was used to assess the model's ability to generalize to previously unseen data.

Three model configurations are compared: Base FP16 (unquantized baseline), Base NF4 (quantized, no fine-tuning), and Fine-tuned NF4 + LoRA. On the original held-out test set (n=150), the fine-tuned model achieves an MAE of 1.35, RMSE of 2.44, exact total-score accuracy of 45.3%, rubric-level accuracy of 85.4%, Pearson correlation of 0.819, and Spearman correlation of 0.853. On the new external test set (n=1,000), the model maintains similar performance with a Pearson correlation of 0.857. These results are promising as a proof-of-concept, though exact score accuracy and error magnitude still leave room for improvement before production use.

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1. Introduction

1.1 Background

In educational institutions, evaluating student responses to open-ended or descriptive questions is a fundamental part of the assessment process. Unlike multiple-choice questions, long-form answers require the evaluator to understand the meaning of the student's response, compare it against a marking scheme or rubric, and assign marks to individual criteria. This process is slow: a teacher grading 100 detailed answers for a single question may spend several hours doing so. Across multiple subjects and batches, this workload becomes difficult to manage consistently.

Manual grading is also inherently subjective. Different evaluators may award different marks for the same answer depending on their understanding, mood, or familiarity with the topic. Even the same evaluator may be inconsistent across a long grading session. These inconsistencies are especially problematic in high-stakes examinations.

Early automated grading approaches relied on keyword matching or surface-level text similarity. These methods fail when students express correct concepts using different words, or when a partially correct answer mixes valid and invalid statements. More recent deep learning methods, particularly Transformer-based language models, can understand semantic meaning and context, making them suitable for grading tasks.

1.2 Problem Statement

The task is to build a system that can automatically assign a numerical score to a long student answer. The system receives a question, a rubric containing multiple grading criteria, and the student's answer. For each criterion in the rubric, the system must decide how many points to award. The sum of criterion-level scores becomes the predicted total score, which is compared against the ground-truth score assigned by a human evaluator.

The main challenge is that answers vary widely in length, vocabulary, and quality. A student may be partially correct, may use different terminology, or may answer only some parts of the question. The grading system must be sensitive to all of these.

1.3 Motivation

Large language models pre-trained on vast text corpora have shown strong ability to follow instructions and reason about text. An instruction-tuned LLM can be prompted to act as a grader, evaluate a student's answer against a rubric, and output a score. However, out-of-the-box LLMs may not follow the exact output format required for automated score parsing, and they may not be calibrated to the scoring standards of a specific dataset. Fine-tuning such a model on a grading dataset teaches it the expected format, the rubric evaluation style, and the scoring distribution — all without changing the base model's general language understanding. QLoRA makes this fine-tuning feasible on a single consumer GPU by reducing the model's memory footprint through 4-bit quantization and training only a small number of additional parameters.

1.4 Proposed System

The proposed system follows a pipeline where the student's answer, the question, and the rubric are assembled into a structured text prompt and passed to the LLM. The model generates a response listing predicted scores for each rubric criterion. These criterion-level scores are parsed, validated against the maximum points, and summed to produce the predicted total score.

Pipeline:

Question + Rubric + Student Answer → Prompt Construction → LLM (Fine-tuned) → Criterion-Level Scores → Score Validation & Clamping → Predicted Total Score → Metric Evaluation

1.5 Scope

This project covers the following: building and formatting a grading dataset of 1,000 records across 10 questions; fine-tuning Qwen2.5-3B-Instruct using QLoRA; evaluating three model configurations (FP16 baseline, NF4 baseline, and fine-tuned NF4+LoRA); and comparing results on two test sets using six quantitative metrics. This is a proof-of-concept research project and is not production-ready.

2. Literature Survey

2.1 Automated Essay Scoring

Automated Essay Scoring (AES) has been an active research area since the 1960s. Project Essay Grade (PEG), developed by Page (1966), used statistical proxies of writing quality. Landauer et al. (1997) introduced Latent Semantic Analysis (LSA) for measuring semantic similarity between student essays and model answers. Foltz et al. (1999) showed that LSA-based systems could achieve inter-rater agreements comparable to human scorers on some tasks.

Machine learning approaches followed, using features such as word counts, syntactic complexity, and vocabulary richness as inputs to regression or classification models. Shermis and Burstein (2013) provide a comprehensive survey of AES methods and challenges, including issues of construct validity and bias in scoring models.

2.2 Transformer-Based Approaches

Devlin et al. (2019) introduced BERT, a pre-trained bidirectional Transformer that significantly improved performance on natural language understanding tasks. Researchers quickly applied BERT to AES, achieving state-of-the-art performance on the Automated Student Assessment Prize (ASAP) dataset. Mayfield and Black (2020) showed that pre-trained language models outperform feature-engineered baselines for short-answer grading tasks.

Yang et al. (2020) proposed using fine-tuned BERT models for rubric-based scoring in educational settings, demonstrating that question-specific fine-tuning helps align model predictions with rubric criteria. Sung et al. (2019) explored Transfer learning-based automated short-answer grading with pre-trained models.

2.3 Large Language Models in Education

The introduction of GPT-3 (Brown et al., 2020) and subsequent instruction-following models demonstrated that LLMs can perform grading tasks in a zero-shot or few-shot manner when prompted appropriately. Kortemeyer (2023) investigated GPT-4's ability to grade physics exam answers and found performance comparable to human teaching assistants for structured problems. Liu et al. (2023) studied LLM-based evaluation of student responses to open-ended science questions and found that GPT-4 with rubric prompts can achieve inter-rater reliability close to human pairs.

However, proprietary models like GPT-4 are expensive to deploy at scale and raise privacy concerns when student data is involved. This motivates the use of smaller, open-weight models that can be hosted locally.

2.4 Parameter-Efficient Fine-Tuning (PEFT)

Hu et al. (2022) introduced LoRA (Low-Rank Adaptation), a method that inserts trainable low-rank matrices into the attention layers of pre-trained models. Instead of updating all model parameters, LoRA trains a small number of adapter parameters while keeping the base model frozen. This reduces GPU memory and training time significantly while achieving performance comparable to full fine-tuning on many tasks.

Dettmers et al. (2023) extended LoRA with 4-bit NF4 quantization to create QLoRA. NF4 quantization maps model weights to a 4-bit normal float format, drastically reducing memory. Double quantization further compresses the quantization constants. QLoRA enables fine-tuning of models with billions of parameters on a single consumer GPU (e.g., a 24 GB VRAM card). The QLoRA paper demonstrated that fine-tuned 7B and 13B models could match or exceed models fine-tuned at FP16 precision on instruction-following benchmarks.

2.5 Qwen2.5 Model Family

The Qwen2.5 series (Qwen Team, 2024) is a family of instruction-tuned language models ranging from 0.5B to 72B parameters. Qwen2.5-3B-Instruct is a 3-billion-parameter model pre-trained on a large multilingual corpus and instruction-tuned using SFT and RLHF. It supports long context windows and follows structured output formats, making it suitable for tasks like rubric-based answer grading where specific output formatting is required.

2.6 Research Gap

While LLMs have been applied to essay scoring and short-answer grading, few studies examine the effect of QLoRA fine-tuning specifically for criterion-by-criterion rubric grading of multi-topic descriptive answers using small open-weight models (under 4B parameters). Existing work either uses large proprietary models (GPT-4) without fine-tuning, or fine-tunes BERT-class models that lack the generative ability to produce structured criterion-level scoring outputs.

This project addresses part of that gap by fine-tuning a 3B instruction-following LLM using QLoRA for criterion-level rubric-based scoring, evaluating it on both a held-out test set from the original data distribution and an entirely new external test set to assess generalization.

3. Objectives

1. Develop an automated grading pipeline for long descriptive student answers using a locally deployable open-weight language model.
2. Implement rubric-level evaluation where each grading criterion is scored independently, with the total score being the sum of criterion scores.
3. Apply QLoRA fine-tuning to adapt Qwen2.5-3B-Instruct to the grading task using a domain-specific dataset of question–answer–rubric triples.
4. Compare the fine-tuned model against two baseline configurations: (a) the unquantized base model in FP16 precision, and (b) the 4-bit NF4 quantized base model without fine-tuning.
5. Evaluate model performance using six quantitative metrics: MAE, RMSE, exact total-score accuracy, rubric-level accuracy, Pearson correlation, and Spearman correlation.
6. Assess model generalization using two separate test evaluations: the original held-out test set (drawn from the same data distribution as training) and a new external test set (collected independently).
7. Identify limitations of the current approach and propose directions for future improvement.

4. Methodology

4.1 Dataset Preparation

The dataset used in this project consists of 1000 student answer records covering 10 questions. Each question comes from a computer science or data engineering topic relevant to machine learning and distributed systems courses. The 10 questions are listed in Table 1.

Table 1: Questions in the Dataset

Question ID	Topic	Question (Abbreviated)
Q01	Gradient Descent Variants	What is Gradient descent? Explain Batch, stochastic and mini batch gradient descent comparatively.
Q02	Overfitting Prevention	Explain the methods of Reducing overfitting in deep learning models shortly.
Q03	Activation Functions	Explain three different activation Functions with pros and cons.
Q04	Transformer Attention	Explain the multihead self and cross attention in the transformer architecture with comparison.
Q05	Apache Kafka	What is Kafka? Explain the different components concisely.
Q06	Hadoop Architecture	Explain the architecture of HDFS, YARN and Map-Reduce in Hadoop.
Q07	Batch Normalisation & Linear Regression	What is Batch Normalisation? Write the 5 assumptions of linear regression.
Q08	NLP Text Preprocessing	Explain five text pre-processing techniques in NLP shortly.
Q09	Ensemble Methods	Explain Bagging, Boosting and Random forest.
Q10	Recurrent Neural Networks	Why we need RNN beyond CNN and ANN?

Each question has exactly 100 student answer records in the complete dataset. All questions carry a maximum score of 10 points, with each question evaluated using a rubric of either 5, 9, or 10 criteria (each worth 1 or 2 points). The rubric structure for the original test set shows 90 records using a 10-criterion rubric, 45 records using a 9-criterion rubric, and 15 records using a 5-criterion rubric. All records in the new external test use a 10-criterion rubric.

Student answers vary in correctness: fully correct, partially correct, incorrect, and irrelevant. The original test set contains 54 fully correct, 42 partially correct, 45 incorrect, and 9 irrelevant answers. This distribution ensures the evaluation covers the full range of student performance.

4.2 Original Dataset Split

The complete dataset of 1000 records was split by question. For each of the 10 questions, 70 records were assigned to training, 15 to validation, and 15 to the original held-out test set. The split was performed with a fixed

random seed of 42 for reproducibility. Records within each split were shuffled after collection to avoid ordering bias.

Table 2: Original Dataset Split

Split	Records	Questions Represented	Purpose
Training	700	Q01–Q10 (70 each)	QLoRA fine-tuning
Validation	150	Q01–Q10 (15 each)	Monitoring training loss
Original Held-Out Test	150	Q01–Q10 (15 each)	Initial evaluation
Total	1000	Q01–Q10	Complete dataset

4.3 New External Test Dataset

After completing the initial round of experiments, a separate new external test dataset of 1000 records was created and evaluated independently. This dataset covers new 10 questions with 100 records per question, but uses a uniform 10-criterion rubric structure for all records. The new external test set was not used during training or validation and represents a strictly held-out evaluation of the model's generalization ability.

The new test set contains questions from different domains like : Operating system, DBMS, Computer Network, Physics, Chemistry (2 questions from each domain).

4.4 Dataset Format

Each record in the dataset is a JSON object with the following fields:

Table 3: Dataset Record Fields

Field	Type	Description
response_id	str	Unique identifier for the student answer (e.g., Q01_S001)
question_id	str	Question identifier (Q01–Q10)
question	str	Full text of the question
student_answer	str	Full text of the student's answer
rubric	list	List of rubric criterion objects
final_score	int/float	Human-assigned total score
max_score	int	Maximum possible score (always 10)
answer_quality	str	One of: fully_correct, partially_correct, incorrect, irrelevant

Each rubric criterion object contains: criterion (str), points (int, maximum for this criterion), points_awarded (int/float, the human-assigned score), rubric_id (str), and label (bool, whether the criterion is satisfied). A representative abbreviated example:

```
{
  "response_id": "Q03_S016",
  "question_id": "Q03",
```

```

    "question": "Explain three different activation Functions with pros and cons.",
    "student_answer": "The sigmoid function is defined as  $\sigma(x) = 1/(1+\exp(-x))$  ...",
    "rubric": [
        {"criterion": "Correctly defines sigmoid function", "points": 1,
        "points_awarded": 1},
        {"criterion": "States at least one pro of sigmoid", "points": 1,
        "points_awarded": 1},
        ...
    ],
    "final_score": 9,
    "max_score": 10
}

```

4.5 Prompt Construction

For both training (JSONL conversion) and inference, prompts are constructed from three components: the question, the rubric (with maximum points per criterion), and the student's answer. The system message instructs the model to act as an expert academic evaluator.

```
prompt = f"""
```

You are an expert academic evaluator.

Your task is to grade a student's answer using the provided question and rubric.

Question:

```
{question}
```

Rubric:

```
{rubric_text}
```

Student Answer:

```
{student_answer}
```

Instructions:

1. Evaluate EVERY rubric criterion independently.
2. Award a score from 0 up to the maximum points specified for that criterion.
3. Never award more points than the maximum points for a criterion.
4. Use only the information in the student's answer when deciding the score.
5. Do not use the expected answer as a separate source of information.
6. Do not provide explanations.
7. Output ONLY one line for each criterion.
8. The output format MUST be:

Criterion 1: [score]

Criterion 2: [score]

Criterion 3: [score]

Continue for every criterion.

Do NOT output a total.

Do NOT output maximum scores.

Do NOT output "/1", "/2", etc.

Do NOT provide explanations.

"""

During inference, only the criterion-level scores are parsed (not the total), and they are validated to ensure they do not exceed the rubric maximum. Any score exceeding the maximum is clamped to the maximum. Missing criteria count as 0 in the predicted total.

4.6 Base Model: Qwen2.5-3B-Instruct

The base model is Qwen2.5-3B-Instruct, part of the Qwen2.5 family from Alibaba Cloud (Qwen Team, 2024). It has approximately 3 billion parameters and was pre-trained on a large multilingual corpus. The instruct variant was further trained with supervised fine-tuning (SFT) and RLHF to follow instructions. Its vocabulary size is 151,643 tokens and it uses a grouped-query attention mechanism. Qwen2.5-3B-Instruct was chosen for this project because it is small enough to fit on a single GPU with 4-bit quantization, yet capable enough to understand structured grading prompts and produce structured output.

4.7 QLoRA Fine-Tuning

QLoRA combines 4-bit quantization with LoRA adapters to enable fine-tuning of large models on limited GPU memory. The following configuration was used:

Table 4: QLoRA Fine-Tuning Configuration

Parameter	Value
Base model	Qwen/Qwen2.5-3B-Instruct
Quantization type	4-bit NF4 (Normal Float 4)
Double quantization	Enabled
Compute dtype	float16
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	0.0002
Epochs	3

Per-device train batch size	1
Gradient accumulation steps	8
Effective batch size	8
Max sequence length	768
Optimizer	paged_adamw_8bit
Warmup steps	20
Weight decay	0.01
Gradient checkpointing	Enabled
Evaluation strategy	Every epoch
Save strategy	Every epoch (best model retained)
Random seed	42
Max new tokens (inference)	300
Decoding strategy	Greedy (do_sample=False)

The LoRA adapters are attached to all seven projection modules in the model's attention and MLP layers. With rank $r=16$ and $\alpha=32$ (scaling factor = $\alpha/r = 2$), the trainable parameter count is a small fraction of the total model parameters (approximately 1–2%), which makes training fast and memory-efficient.

Training completed in 37.5 minutes. The final training loss was 0.2531 and the validation loss was 0.1006. The best model checkpoint (by validation loss) was loaded at the end of training.

4.8 Training Algorithm

Algorithm 1: QLoRA Fine-Tuning for Automated Answer Grading

Algorithm 1: QLoRA Fine-Tuning

Input: Base model M (Qwen2.5-3B-Instruct), Training dataset D_{train} (700 records), Validation dataset D_{val} (150 records), LoRA config ($r=16$, $\alpha=32$, dropout=0.05)

1. Load M from HuggingFace Hub.
2. Apply 4-bit NF4 quantization with double quantization enabled. Compute dtype = float16.
3. Call `prepare_model_for_kbit_training(M)` to enable gradient checkpointing and prepare quantized layers.
4. Attach LoRA adapters to: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. Freeze all base model parameters.
5. Format D_{train} and D_{val} as JSONL with system/user/assistant message structure.
6. Initialize SFTTrainer with `paged_adamw_8bit`, `lr=2e-4`, `warmup_steps=20`, `max_length=768`, `gradient_accumulation=8`.
7. For each of 3 epochs: compute forward pass on batch → compute cross-entropy loss on assistant tokens → backpropagate through LoRA parameters only → update using optimizer.
8. After each epoch: evaluate on D_{val} , log validation loss. Save checkpoint if validation loss improves.
9. Load best checkpoint. Save LoRA adapter weights to disk as `final_adapter`.

Output: Trained LoRA adapter ready for inference.

4.9 Inference Algorithm

Algorithm 2: Inference and Scoring

Input: Test record with question, student_answer, rubric; base model M; LoRA adapter.

1. Load M with 4-bit NF4 quantization. Load LoRA adapter via PeftModel. Set model to eval() mode.
2. Load tokenizer. Set pad_token = eos_token if necessary.
3. Build grading prompt: combine question, rubric (with maximum points per criterion), and student answer.
4. Apply chat template. Tokenize prompt.
5. Run model.generate() with greedy decoding (do_sample=False), max_new_tokens=300.
6. Decode generated tokens (excluding prompt tokens) to get response string.
7. Parse response with regex: extract criterion number and score from each line.
8. Validate each predicted score: clamp negative scores to 0; clamp scores exceeding rubric maximum to that maximum. Track missing criteria and invalid predictions.
9. Sum validated criterion scores to produce predicted total.
10. Compare predicted total to ground-truth final_score. Compute absolute error and rubric-level accuracy.

Output: Predicted total score, per-criterion scores, absolute error, rubric accuracy.

4.10 Baseline Evaluation

Three model configurations are compared to isolate the contribution of each component:

Table 5: Model Configurations

Configuration	Precision	LoRA Adapter	Key Comparison
Base FP16	float16 (no quantization)	No	Upper-bound unquantized reference
Base NF4	4-bit NF4	No	Shows quantization-only effect
Fine-tuned NF4 + LoRA	4-bit NF4	Yes (r=16)	Primary experiment

The most controlled comparison is Base NF4 vs Fine-tuned NF4 + LoRA, because both use the same quantized base model and differ only in whether the fine-tuned LoRA adapter is present. The FP16 baseline serves as an additional reference showing the effect of quantization on the untuned model.

4.11 Evaluation Metrics

Six metrics are used to evaluate each model configuration:

Mean Absolute Error (MAE): $MAE = (1/N) \times \sum |predicted_i - actual_i|$. Average number of points by which the model's prediction deviates from the human score. Lower is better. An MAE of 1.0 means the model is off by one point on average.

Root Mean Square Error (RMSE): $RMSE = \sqrt{(1/N) \times \sum (\text{predicted}_i - \text{actual}_i)^2}$. Similar to MAE but penalizes large errors more heavily due to the squaring. Lower is better.

Exact Total-Score Accuracy: $Accuracy = (1/N) \times \sum 1[\text{predicted}_i == \text{actual}_i]$. Fraction of test records where the model's predicted total exactly matches the human-assigned total. Higher is better.

Rubric-Level Accuracy: $Rubric\ Acc. = (\sum \text{correct criteria}) / (\sum \text{evaluatable criteria})$. Fraction of individual criterion-level predictions that exactly match the human-assigned criterion score. Higher is better.

Pearson Correlation (r): $r = \sum [(x - \bar{x})(y - \bar{y})] / (\sum (x - \bar{x})^2 \times \sum (y - \bar{y})^2)^{0.5}$. Measures linear association between predicted and actual scores. Ranges from -1 to 1. Higher absolute value is better.

Spearman Correlation (ρ): $\rho = \text{Pearson}(\text{rank}(\text{predicted}), \text{rank}(\text{actual}))$. Measures rank correlation between predicted and actual scores, less sensitive to outliers. Ranges from -1 to 1.

4.12 Time and Space Complexity

Let N = number of training examples, L = input sequence length (max 768 tokens), G = maximum generated tokens (300), P = total model parameters (~3B), R = LoRA rank (16), and M = number of LoRA target modules (7).

Inference time complexity: Each forward pass through the Transformer is $O(L^2 \times d)$ per layer (attention) and $O(L \times d_{ff})$ per layer (MLP), where d is the hidden dimension and d_{ff} is the feed-forward dimension. Autoregressive generation of G tokens requires G forward passes. With the KV-cache, the per-step cost after the first token is $O(L_{cached} \times d)$ per layer rather than $O(L^2 \times d)$. Practically, inference for one record takes a few seconds on GPU.

Training time complexity: Full fine-tuning would update all P parameters each step: $O(N \times L^2 \times P)$. With LoRA, only $2 \times R \times d$ parameters per target module are updated (for $M = 7$ modules, this is $2 \times 7 \times 16 \times d$ trainable parameters per layer). The gradient computation is still $O(L^2 \times d)$ per layer, but the parameter update step is $O(R \times d)$ rather than $O(d^2)$, saving substantial computation. Gradient checkpointing trades off compute for memory by recomputing activations during the backward pass instead of storing them.

Memory: 4-bit NF4 quantization reduces model weights from 16-bit to 4-bit, cutting memory by approximately 4×. The base model at FP16 requires ~6 GB; at NF4 it requires approximately 2–3 GB, leaving headroom for activations and optimizer states. This is what makes training feasible on a single GPU.

In practice, training 3 epochs on 700 examples took approximately 37.5 minutes (2253 seconds).

5. Experimental Setup

The following hardware and software environment was used for all experiments. GPU information is reported from the training scripts' output.

Table 6: Experimental Setup

Parameter	Value
Model	Qwen/Qwen2.5-3B-Instruct
Framework	PyTorch (CUDA)
Fine-tuning library	HuggingFace Transformers + PEFT + TRL (SFTTrainer)
Quantization	bitsandbytes (4-bit NF4)
Training dataset size	700 records
Validation dataset size	150 records
Original held-out test size	150 records
New external test size	1000 records
Number of questions	10
Max sequence length	768
Training epochs	3
Effective batch size	8
Learning rate	0.0002
Optimizer	paged_adamw_8bit
Decoding (inference)	Greedy (do_sample=False), max_new_tokens=300
Random seed	42
Training time	2253 seconds (~37.5 minutes)
Final training loss	0.2531
Final validation loss	0.1006
Validation token accuracy	97.34%

Note: Specific GPU model and VRAM were not logged to a file. The training scripts print GPU information at runtime. The max sequence length of 768 was selected based on token length analysis showing that the majority of training examples fit within this limit, balancing coverage and GPU memory usage.

6. Experimental Results and Discussion

6.1 Experiment A: Original Held-Out Test Set (n=150)

The original held-out test set contains 150 records drawn from the same dataset distribution as the training and validation sets. Table 7 shows the results for all three model configurations.

Table 7: Results on Original Held-Out Test Set (n=150)

Model	MAE↓	RMSE↓	Exact Acc.↑	Rubric Acc.↑	Pearson r↑	Spearman ρ↑
Base FP16	2.2333	3.3397	23.33%	72.61%	0.6785	0.6534
Base NF4	2.4600	3.5637	22.67%	71.74%	0.6369	0.6146
Fine-tuned (QLoRA)	1.3467	2.4413	45.33%	85.44%	0.8186	0.8527

The fine-tuned model is clearly the best performer across all six metrics on the original held-out test set:

- MAE reduced from 2.4600 (Base NF4) to 1.3467 — a 45.3% improvement.
- Pearson correlation improved from 0.6369 to 0.8186.
- Exact total-score accuracy roughly doubled from 22.67% to 45.33%.

Comparing the two baselines: Base NF4 is slightly worse than Base FP16 across most metrics, confirming that 4-bit quantization alone (without fine-tuning) degrades performance. However, adding the LoRA adapter on top of the NF4 base more than compensates, achieving results well above the FP16 baseline.

The following figures show individual metric comparisons for the original held-out test set.

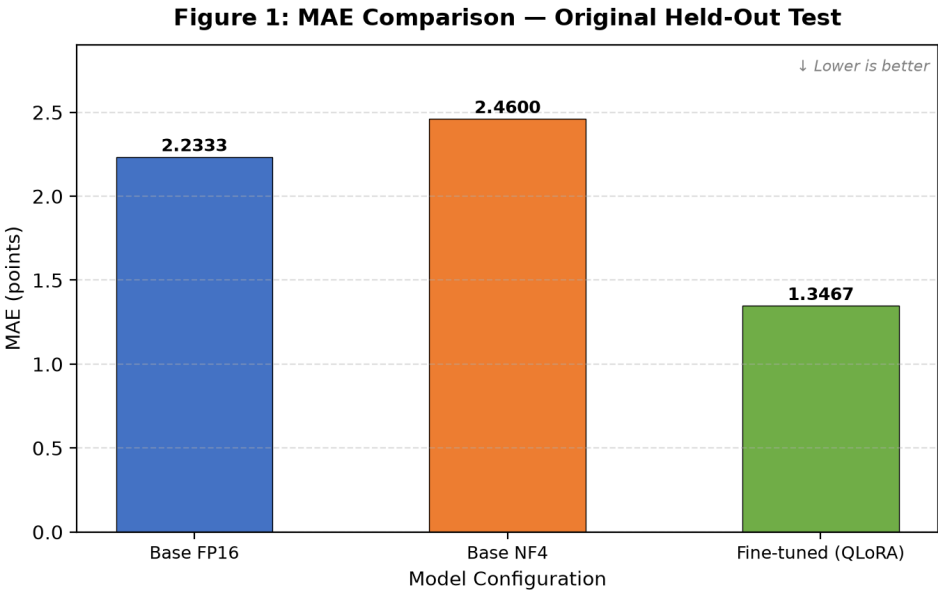


Figure 1: MAE Comparison — Original Held-Out Test Set (lower is better)

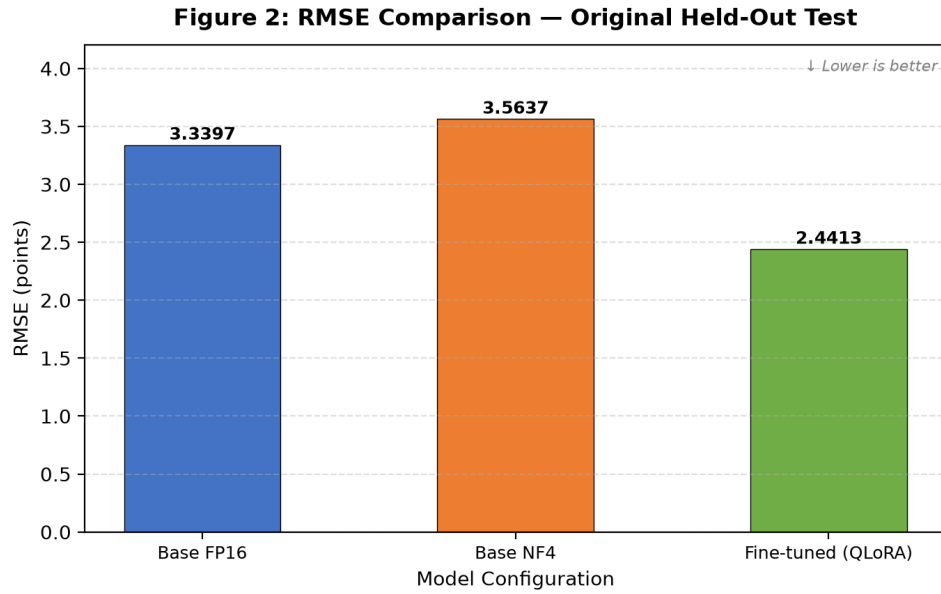


Figure 2: RMSE Comparison — Original Held-Out Test Set (lower is better)

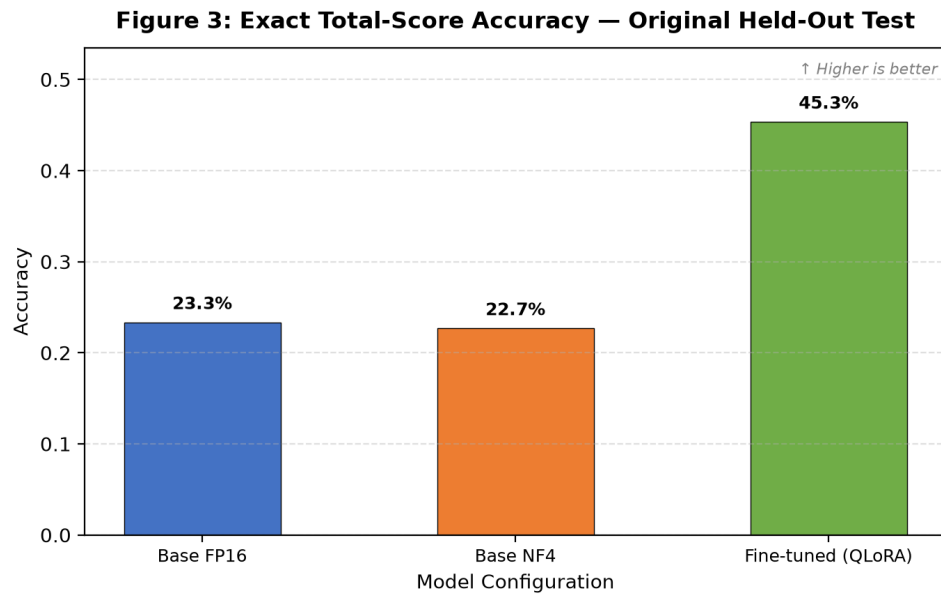


Figure 3: Exact Total-Score Accuracy — Original Held-Out Test Set (higher is better)

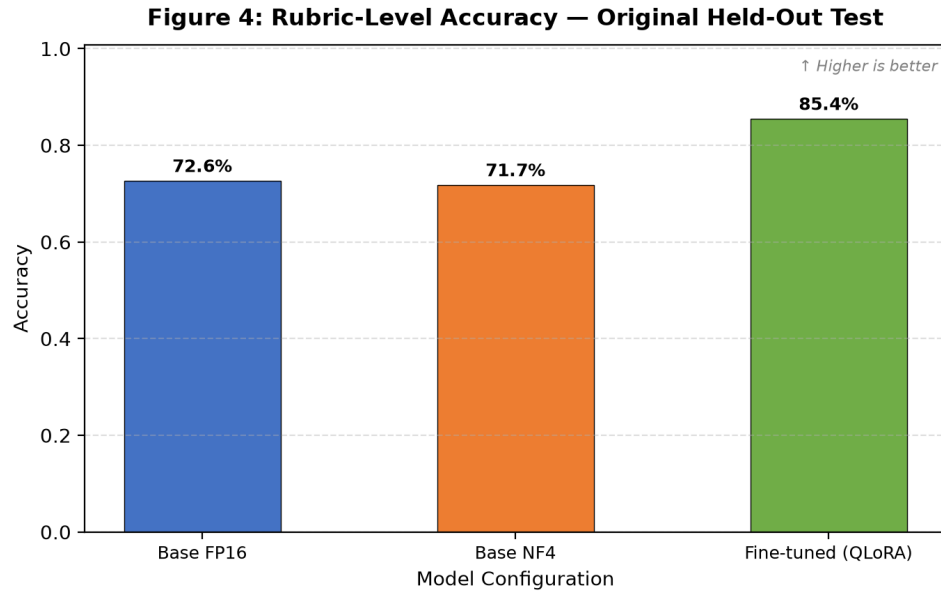


Figure 4: Rubric-Level Accuracy — Original Held-Out Test Set (higher is better)

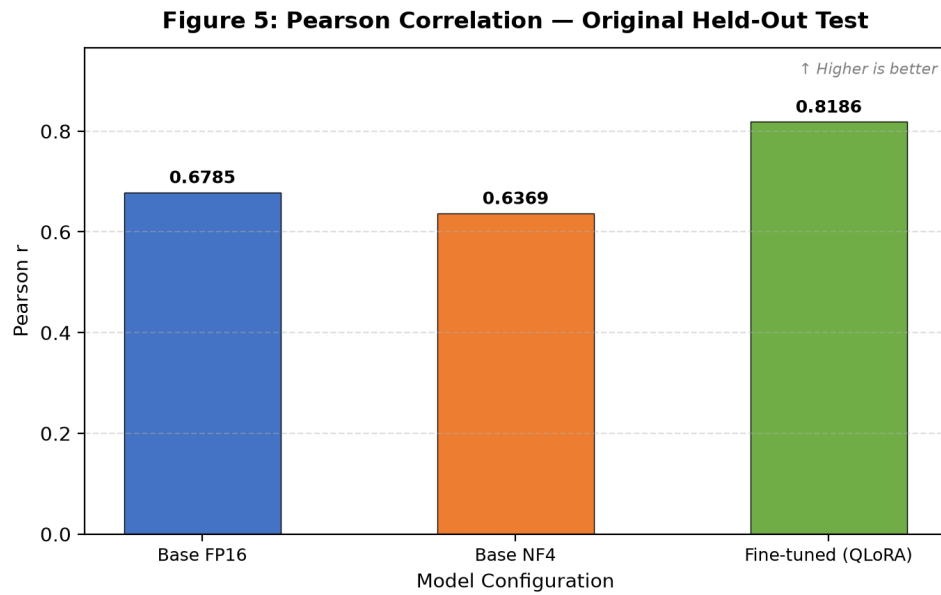


Figure 5: Pearson Correlation — Original Held-Out Test Set (higher is better)

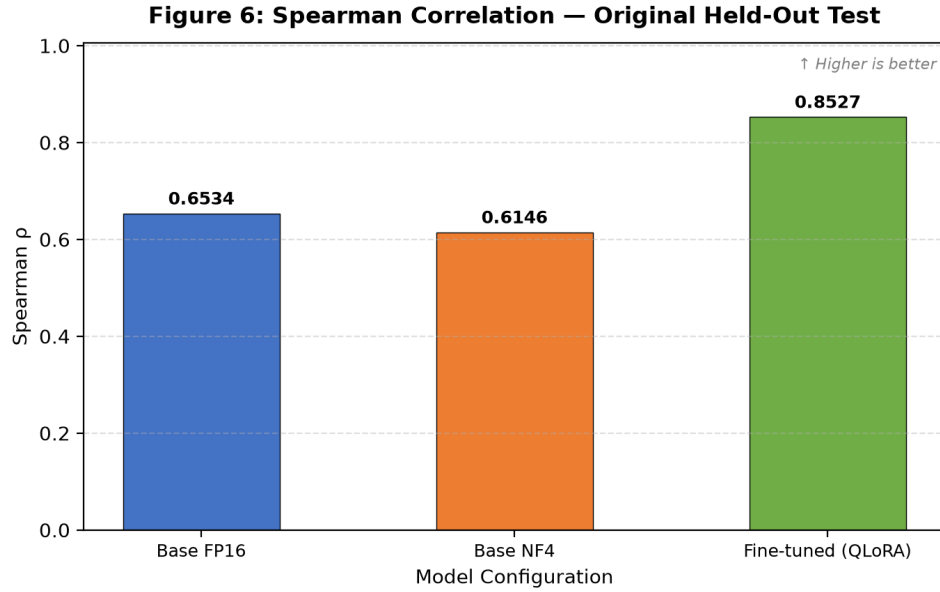


Figure 6: Spearman Correlation — Original Held-Out Test Set (higher is better)

Figure 8: Actual vs. Predicted Score Scatter — Fine-Tuned Model (Original Test)

6.2 Experiment B: New External Test Set (n=1,000)

The new external test set contains 1000 records covering the new 10 questions but using a uniform 10-criterion rubric structure. This test was run after the original experiments to evaluate whether the model generalizes to answers outside its training distribution. Table 8 shows the results.

Table 8: Results on New External Test Set (n=1,000)

Model	MAE↓	RMSE↓	Exact Acc.↑	Rubric Acc.↑	Pearson r ↑	Spearman ρ ↑
Base FP16	2.3880	3.2168	22.60%	73.82%	0.7272	0.7284
Base NF4	2.9920	3.9764	21.10%	69.54%	0.6666	0.6632
Fine-tuned (QLoRA)	1.3790	2.2034	38.90%	83.77%	0.8570	0.8534

On the new external test set, the fine-tuned model again outperforms both baselines by a substantial margin:

- MAE: 1.3790 vs 2.3880 (Base FP16) and 2.9920 (Base NF4).
- Pearson correlation: 0.8570 — higher than on the original test (0.8186).
- Exact accuracy dropped from 45.33% (original test) to 38.90% (new test), which is expected as the new test set has 1,000 records with a different score distribution, making exact matches inherently harder.

Notably, both baselines also improve their Pearson correlation on the new test set, likely because the new test set's score distribution is more balanced, which aids correlation measurement. The Base NF4 model's MAE worsens on the new test set (2.9920 vs 2.4600 on original), reflecting the difficulty of the new test.

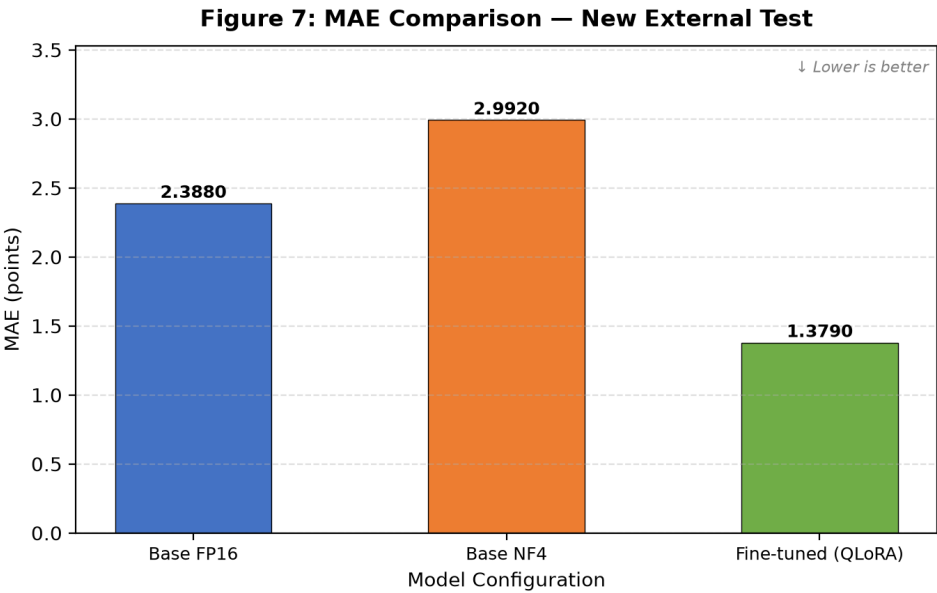


Figure 9: MAE Comparison — New External Test Set (lower is better)

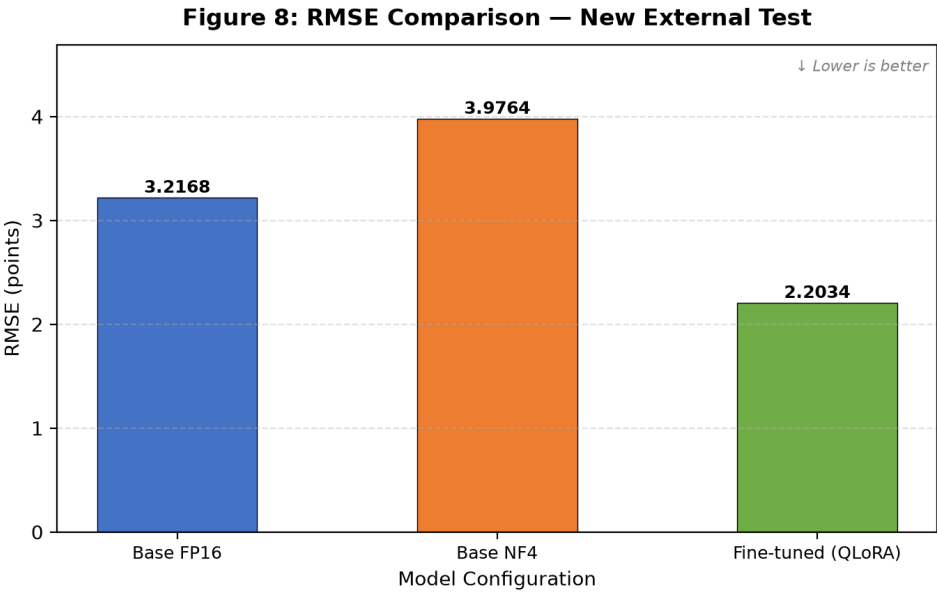


Figure 10: RMSE Comparison — New External Test Set (lower is better)

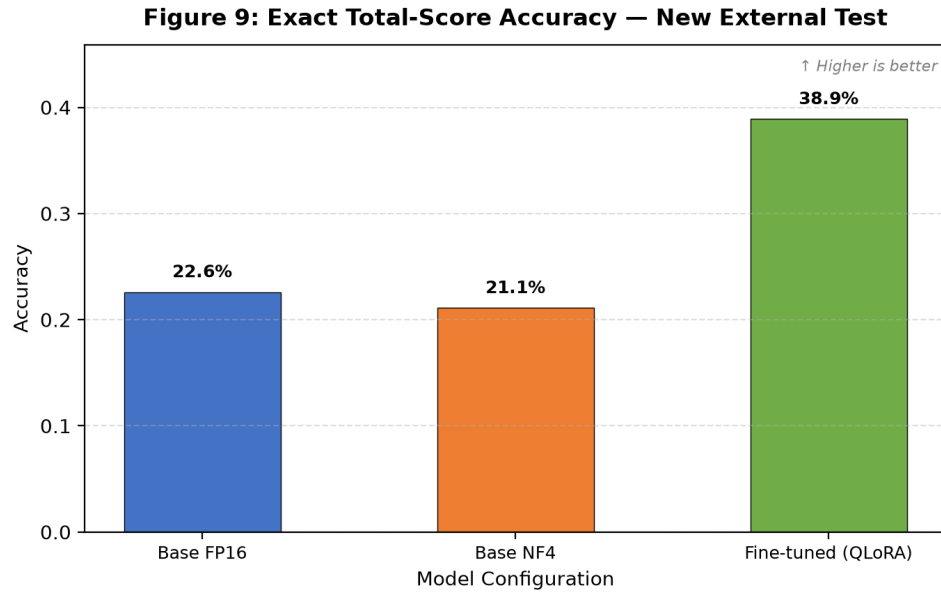


Figure 11: Exact Total-Score Accuracy — New External Test Set (higher is better)

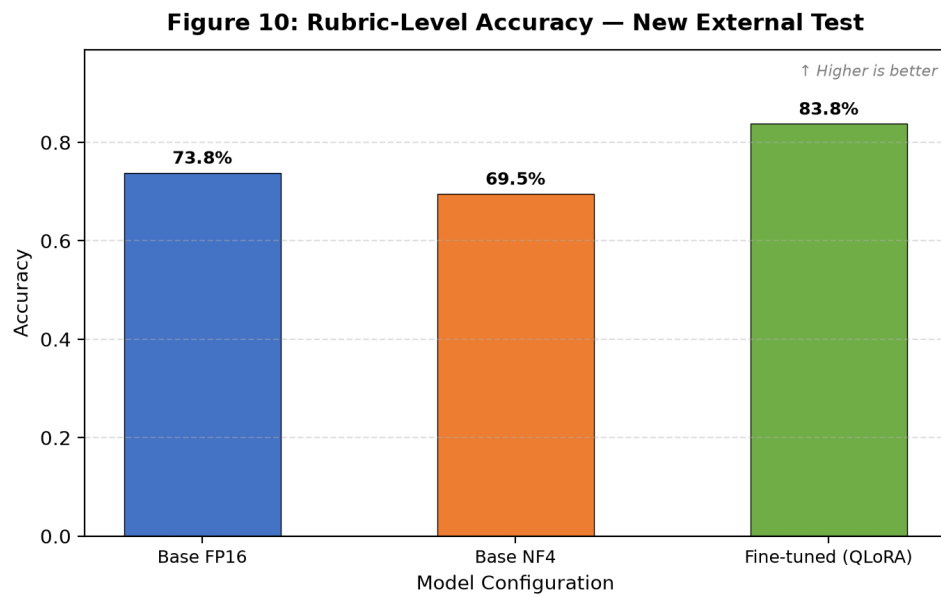


Figure 12: Rubric-Level Accuracy — New External Test Set (higher is better)

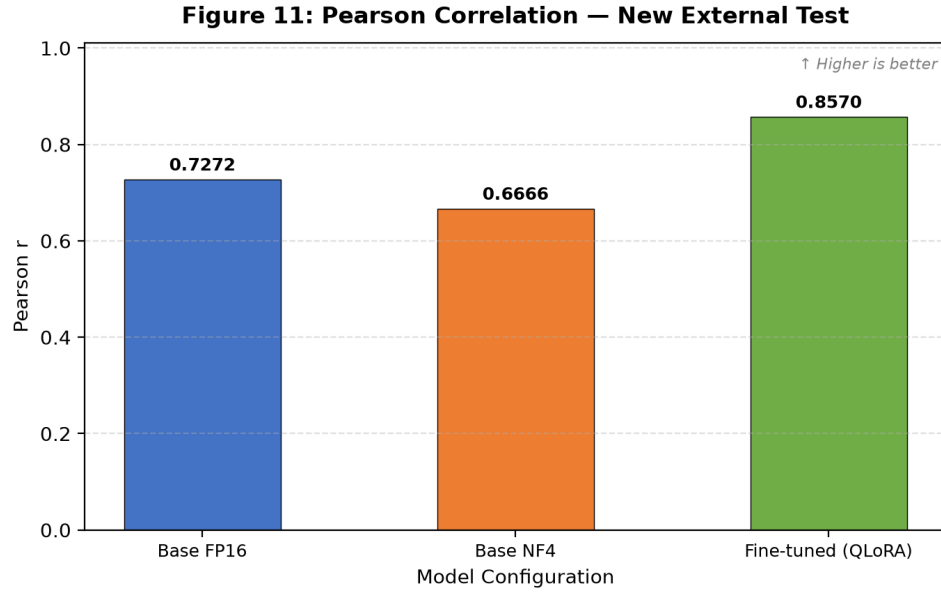


Figure 13: Pearson Correlation — New External Test Set (higher is better)

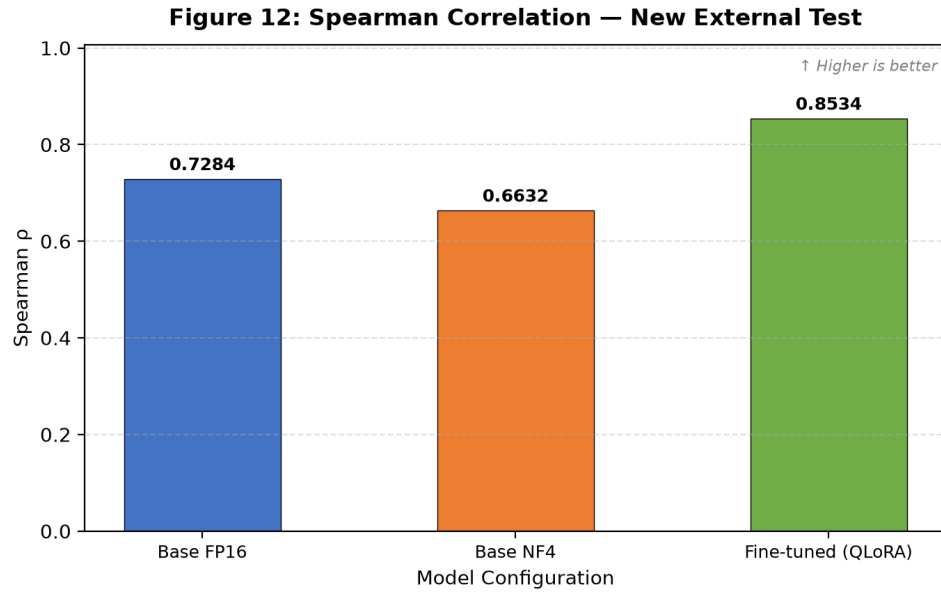


Figure 14: Spearman Correlation — New External Test Set (higher is better)

6.3 Cross-Test Comparison

Table 9 consolidates results from both test sets to facilitate direct comparison of each model's performance across the two evaluation scenarios.

Table 9: Cross-Test Comparison — All Models, Both Test Sets

Model	Test Set	MAE↓	RMSE↓	Exact Acc.↑	Rubric Acc.↑	Pearson↑	Spearman↑
Base FP16	Original	2.2333	3.3397	23.33%	72.61%	0.6785	0.6534
Base FP16	New Ext.	2.3880	3.2168	22.60%	73.82%	0.7272	0.7284
Base NF4	Original	2.4600	3.5637	22.67%	71.74%	0.6369	0.6146
Base NF4	New Ext.	2.9920	3.9764	21.10%	69.54%	0.6666	0.6632
Fine-tuned (QLoRA)	Original	1.3467	2.4413	45.33%	85.44%	0.8186	0.8527
Fine-tuned (QLoRA)	New Ext.	1.3790	2.2034	38.90%	83.77%	0.8570	0.8534

The cross-test comparison reveals several important observations:

1. The fine-tuned model maintains strong performance on the new external test. Its MAE barely changes (1.3467 → 1.3790) and its Pearson correlation slightly improves (0.8186 → 0.8570), suggesting the fine-tuning generalized reasonably well to the new test rather than overfitting to the original training distribution.
2. Base NF4 degrades more on the new external test (MAE: 2.46 → 2.99) than Base FP16 does (MAE: 2.23 → 2.39), indicating that quantization without fine-tuning reduces robustness to distribution shifts.
3. Exact accuracy consistently decreases when moving to the new external test for all models. This is partly because the new test has 1,000 records with a different and more uniform score distribution, which is inherently harder to predict exactly.
4. Both correlation metrics (Pearson and Spearman) improve for all models on the new external test, which may reflect the more balanced score distribution in that dataset enabling better rank discrimination.

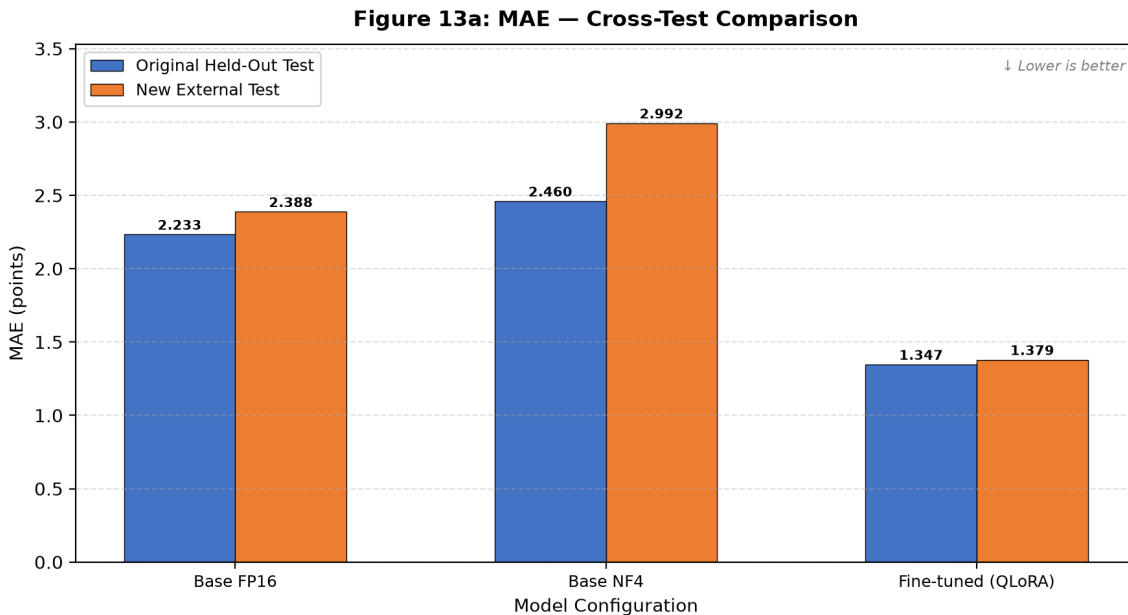


Figure 15a: MAE Cross-Test Comparison — Original vs New External

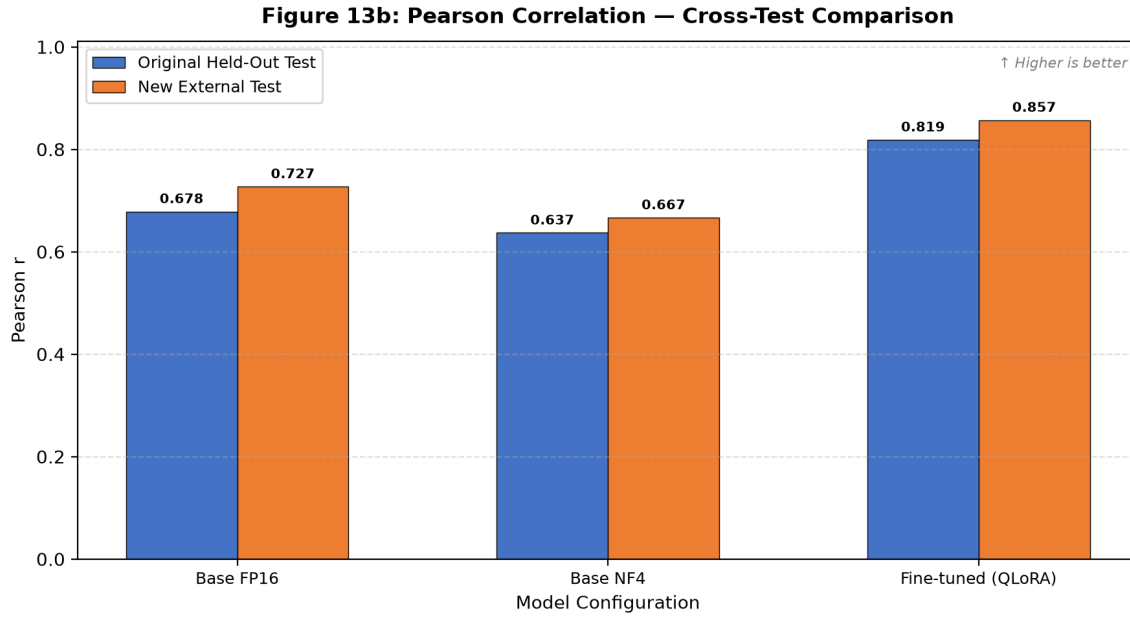


Figure 15b: Pearson Correlation Cross-Test Comparison

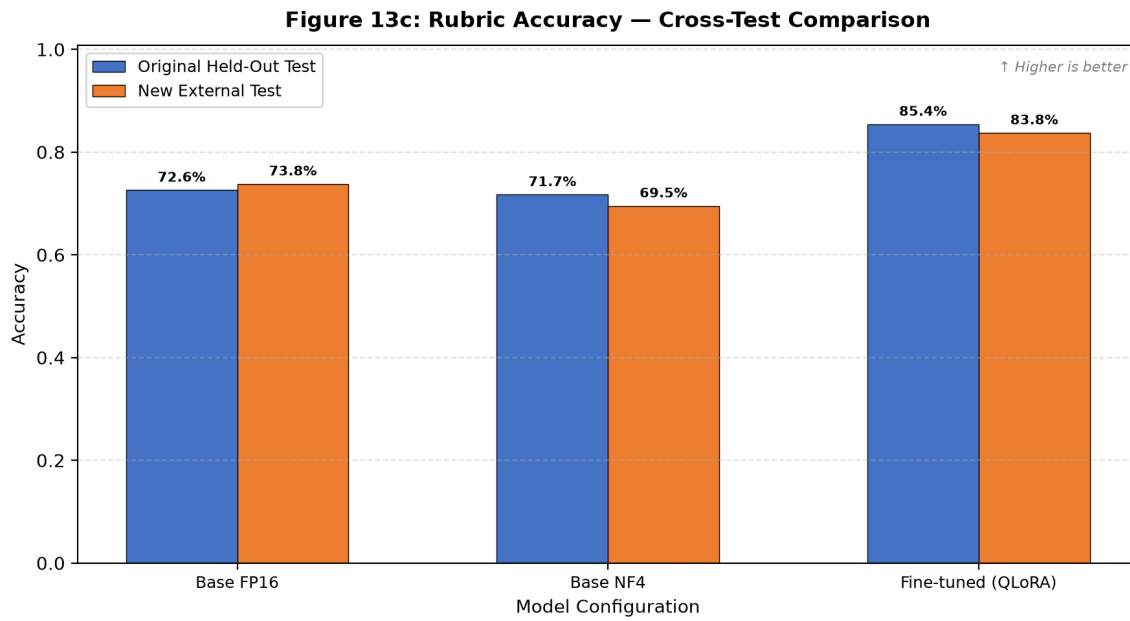


Figure 15c: Rubric Accuracy Cross-Test Comparison

6.4 Error Analysis

An analysis of the fine-tuned model's predictions on the original held-out test set (n=150) reveals the following error distribution:

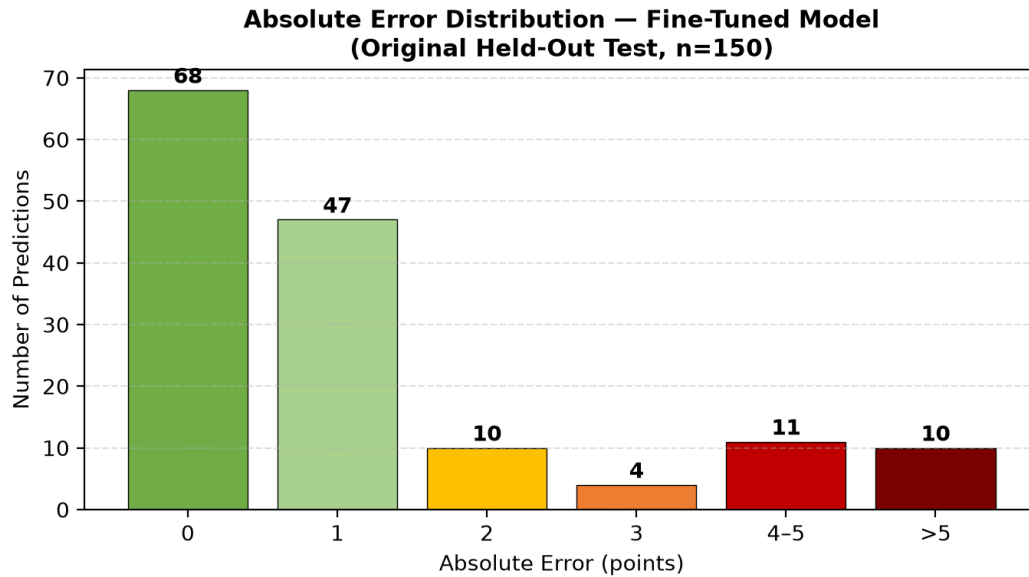


Figure 16: Absolute Error Distribution — Fine-Tuned Model (Original Held-Out Test)

Key observations from the error distribution:

- 68 out of 150 predictions (45.3%) are exactly correct.
- 115 predictions (76.7%) have an error of 1 point or less.
- 125 predictions (83.3%) have an error of 2 points or less.
- 21 predictions (14.0%) have an error greater than 3 points, including a few extreme cases with errors of 8 or 9 points.
- 66 predictions underestimate the score; 16 overestimate; 68 are exact.
- 6 records had one or more missing criteria (the model did not output a score for that criterion). Missing criteria were counted as 0.

Table 10 shows representative prediction examples from the original held-out test set:

Table 10: Representative Error Analysis Examples

Case Type	Response ID	Actual Score	Predicted Score	Observation
Exact Match	Q03_S029	1 / 10	1 / 10	Low-scoring answer; model correctly identifies all criteria as unmet.
Overestimation (+8)	Q01_S005	2 / 10	10 / 10	Severe overestimation; model may have been misled by confident but incorrect answer.
Underestimation (-9)	Q04_S075	9 / 10	0 / 10	Complete failure; model failed to parse or assign scores for this

				answer.
Near-Miss (−1)	Q09_S061	7 / 10	6 / 10	One criterion incorrectly scored; typical small-error case.

The extreme cases (± 8 or ± 9 errors) suggest occasional model failures where either the model awards full marks to a weak answer or fails to recognize a strong answer. These edge cases may stem from prompt sensitivity, unusual answer styles, or ambiguous rubric criteria. The majority of predictions, however, fall within 1–2 points of the ground truth.

6.5 Result Interpretation

The following summarizes the key findings and their interpretation:

Lowest MAE: Fine-tuned (QLoRA) — 1.3467 (original test), 1.3790 (new test).

Lowest RMSE: Fine-tuned (QLoRA) — 2.4413 (original), 2.2034 (new).

Highest exact accuracy: Fine-tuned (QLoRA) — 45.33% (original), 38.90% (new). Both baselines are around 22–23%.

Highest rubric-level accuracy: Fine-tuned (QLoRA) — 85.44% (original), 83.77% (new).

Highest Pearson r: Fine-tuned (QLoRA) — 0.8186 (original), 0.8570 (new).

Effect of NF4 quantization on the base model: Comparing Base FP16 and Base NF4, the quantized model is consistently slightly worse (higher MAE and lower correlation), confirming that quantization alone reduces grading quality.

Effect of LoRA fine-tuning: Adding the LoRA adapter to the NF4 base model dramatically improves all metrics. The MAE improvement vs Base NF4 is approximately 45% on the original test. The Pearson correlation improves from 0.637 to 0.819 — a substantial gain.

6.6 Is the Result Good?

The fine-tuned model's results are promising for a research prototype built with a 3B parameter model and a training set of only 700 examples. However, an honest assessment must acknowledge both the strengths and the weaknesses.

STRENGTHS:

- A Pearson correlation of 0.82–0.86 is strong and indicates the model's scoring is meaningfully aligned with human scoring across the full score range.
- Rubric-level accuracy of 83–85% means the model correctly assigns individual criterion scores most of the time.
- The model generalizes well: its MAE and correlation are nearly the same on the new external test as on the original held-out test, despite the larger size and different composition of the new test.
- The improvement over both baselines is large and consistent across all metrics, confirming that fine-tuning genuinely helps.

WEAKNESSES:

- Exact total-score accuracy of 38.9–45.3% means that more than half of all predictions miss the exact total score, even if only by 1 point. In a high-stakes examination context, this level of discrepancy would require human review.
- The MAE of 1.35–1.38 points (out of 10) may be acceptable for screening purposes but is too large for final grade assignment without human verification.
- A small fraction of predictions have very large errors (≥ 5 points), which is potentially harmful in an automated system without safeguards.
- The training dataset is relatively small (700 records) and limited to 10 questions in one domain (computer science).

VERDICT: The result is encouraging and represents a valid proof-of-concept, but it is not yet good enough for unsupervised deployment in real grading scenarios. Larger datasets, more diverse questions, and human-in-the-loop validation would be necessary before practical use.

6.7 Limitations

The following limitations apply to the current project:

1. Small training dataset: With only 700 training records across 10 questions, the model has limited exposure to the variety of student writing styles, misconceptions, and subject topics it would encounter in a real examination system.
2. Limited question and subject diversity: All 10 questions are from computer science and data engineering. The model has not been evaluated on other subjects or question types.
3. Synthetic/generated aspects: It is unclear from the project files how the student answers were collected. If answers were synthetically generated, they may not reflect the full range of real student writing.
4. No inter-rater agreement measurement: The ground-truth scores are assigned by human evaluators, but the agreement between different human evaluators is not reported. Without knowing the human-human agreement level, it is difficult to judge whether the model's correlation of 0.82 is close to human performance.
5. No calibration: The model can produce extreme scores in some cases (error > 5). Calibration techniques or score-range constraints could reduce these outliers.
6. Prompt sensitivity: Results may vary with different prompt formulations. The current prompt was not systematically optimized.
7. Single model family: Only Qwen2.5-3B-Instruct was evaluated. Other models of similar size (e.g., Llama-3.2-3B, Phi-3.5-mini) may perform differently.

7. Conclusion

This project addressed the problem of automatically grading long descriptive student answers using a large language model fine-tuned with QLoRA. The base model, Qwen2.5-3B-Instruct, was adapted to the grading task by training a lightweight LoRA adapter (rank 16) on a dataset of 700 question–answer–rubric triples across 10 computer science questions.

On the original held-out test set (150 records), the fine-tuned model achieved an MAE of 1.3467, exact score accuracy of 45.33%, rubric-level accuracy of 85.44%, and Pearson correlation of 0.8186. On the new external test set (1,000 records), performance was similar: MAE 1.3790, Pearson 0.8570.

QLoRA fine-tuning clearly improved performance relative to both baselines (unquantized FP16 and 4-bit NF4 without fine-tuning), confirming that task-specific adaptation is necessary for good grading performance. The model generalizes reasonably well to the new external test set, showing that the fine-tuning did not overfit to the original training distribution.

However, the results are not strong enough for deployment without human oversight. Exact score accuracy remains below 50%, and a small fraction of predictions have large errors that could harm students if taken at face value. The current system is best understood as a proof-of-concept demonstrating that a small, locally-hosted LLM fine-tuned with QLoRA can produce grading correlations in the range of 0.82–0.86 — which is a useful starting point, but not a final product.

The primary recommendation for improvement is to collect more training data across a wider range of subjects and question types, combined with a larger base model and human-in-the-loop validation for extreme predictions.

8. Future Work

8. Collect a larger and more diverse training dataset spanning multiple subjects, question types, and student populations. At least 5,000–10,000 records across 50+ questions would substantially improve generalization.
9. Evaluate larger base models (7B, 14B parameters) to determine whether model scale provides meaningful gains in grading accuracy.
10. Conduct systematic prompt engineering or prompt optimization to reduce the frequency of extreme prediction errors.
11. Implement score calibration: post-process model predictions to reduce variance and eliminate extreme outliers.
12. Add a confidence or uncertainty estimate so that predictions with high uncertainty can be flagged for human review.
13. Compare QLoRA fine-tuning with other PEFT methods (prompt tuning, prefix tuning, IA³) to identify the most efficient adaptation approach.
14. Evaluate on real student answers collected in a controlled examination setting, rather than answers that may have been generated or semi-synthetic.
15. Measure inter-rater agreement among human evaluators to establish a realistic performance ceiling for the automated system.
16. Explore Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO) to align model scoring more closely with human evaluator preferences.
17. Extend to cross-subject evaluation: train on one set of subjects and test on a different set to measure domain transfer ability.

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